

Project Design - Solution Architecture

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	4 Marks

Solution Architecture Diagram

The system architecture diagram below illustrates the components, processes, data flow, and technology layers of the **Book a Doctor** platform.

```

graph TD
    subgraph Client_Tier [React Frontend]
        UI[Bootstrap 5 Glassmorphic UI]
        Context[Auth & Theme Context]
        AxiosInst[Axios HTTP Client]
    end

    subgraph Logic_Tier [Node/Express API]
        Router[Express Router]
        AuthGuard[JWT Auth Guard Middleware]
        RoleCheck[RBAC Middleware]
        MulterFile[Multer File Upload Validator]
        Controller[Controllers: Auth, Appt, Doctor, Admin]
    end

    subgraph Data_Tier [Data Storage]
        DB[(MongoDB Atlas Database)]
        Disk[Local uploads/ Storage Directory]
    end

    UI --> Context
    UI --> AxiosInst
    AxiosInst -- HTTPS REST Requests --> Router
    Router --> AuthGuard
    AuthGuard --> RoleCheck
    RoleCheck --> MulterFile
    MulterFile --> Controller

    Controller -->|Query / Save| DB
    Controller -->|Write file to disk| Disk

```

Key Architecture Integrations

1. Security Integration (JWT + RBAC)

All requests targeting `/api/doctor/*` or `/api/admin/*` must pass through the `authGuard` middleware which reads JWT from headers or `HttpOnly` cookies. If valid, the `roleCheck` middleware verifies if the user's role matches the requested resource.

2. Multer Locker Integration

When a patient uploads a record via `/api/patient/upload`, the file goes through Multer file filters which validates file extension and size, writing it to `backend/uploads/` and recording the filepath metadata in the `Report` database collection.